Static Voltage Stability Based Maximum Integration Ratio of Distributed Renewable Energy in Distribution Power System

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Abstract. **Distributed renewable energy source (DREs) is an effective supplement to traditional centralized power generation systems. However, how to allocate the DREs and how to evaluate the maximum integration ratio of DREs are the two main obstacles hindering the development of DREs. To tackle these problems, this paper proposes a novel optimal allocation method for DREs in distribution power systems with static voltage stability as the constraint. Based on the optimal allocation method, one maximum integration ratio of DREs in the distribution power system is studied. The theoretical analysis and case studies show the effectiveness of the proposed allocation and evaluation methods of DREs. Moreover, the proposed allocation method of DREs can achieve high efficiency and low power loss.**

Keywords: distributed renewable energy source;static voltage stability;distribution power system;optimal allocation method

I. Introduction

Distributed renewable energy source (DREs) is an effective supplement to traditional centralized power generation systems. The combined and complementary power generation model of traditional centralized power generation and DREs has the functions of achieving green development, reducing transmission losses, saving investment, and improving system reliability. It breaks the shortcomings of high pollution and low reliability of traditional centralized power supply systems, improves energy utilization efficiency, and flexibly distributes location-allocation, satisfying the characteristics of existing resource distribution and the demand for decentralized power supply [1-2].

However, the dynamic reactive power support capability of renewable energy units is weak [3]. With the rapid increase in the proportion of renewable energy, the system voltage support capability drops sharply, and the problem of voltage stability is prominent. The static voltage stability can reflect the voltage stability level of the system to a certain extent. Therefore, studying the static voltage stability and its margin of the new energy multi-station transmission system is of great significance for the safe and stable operation of the system [4].

One of the main factors limiting the DREs level is the voltage exceeding the limit caused by the reverse power flow caused by the renewable energy access. A simple solution is to reduce the voltage setting of the substation or use an automatic voltage controller (AVC) on the feeder. However, the access capacity and location of DREs are different, so the allocation of AVC or dynamic reactive power compensator is complicated. Besides, the investment of the extra AVC equipment in each accession node is high. Moreover, the voltage distribution of the whole feeder cannot be guaranteed in an acceptable range by just the on-load tap-changer due to the dynamic changes in the renewable energy output. In [5], an approach to find suitable sites for integrating DREs on the basis of maximum power that can be injected at these sites without violating transmission network constraints is proposed, but the approach is a little complicated. In [6], a technique for finding the optimal locations and sizes of (RESs) in electric power systems is proposed by using oscillatory particle swarm optimization. However, the static voltage constraint is not fully considered in these available studies.

Therefore, this paper intends to study the maximum integration of DREs in the distribution power system while considering static voltage stability. The main contributions of this paper are as follows:

1) the optimal allocation method of DREs in the distribution power system is proposed.

2) the maximum integration ratio of DREs with static voltage stability as a constraint is proposed.

II. Static voltage stability of distribution power system

The basic unit of the distribution power system is the feeder. The first end of the feeder is connected to the high-voltage distribution network through a high-voltage step-down transformer, and the end is connected to the user through a low-voltage step-down transformer. The voltage level of most feeder lines in China is 10 kV, and the lines on each feeder are distributed in a tree shape, connected to several distribution transformers through a radial network.

A. Static voltage stability with one DREs

As shown in Figure 1, U_0 is the voltage of the source node, representing the low voltage bus of the substation; U_1 is the voltage of Bus 1, representing the power load node; $P_1 + jQ$ is the power load, and $R_1 + jX_1$ is the equivalent impedance of the distribution line; $P_{dg1} + jQ_{dg1}$ is the DREs power integrated into Bus 1. The static voltage stability limit of a node (hereinafter referred to as the "static stability limit") refers to the critical value at which the slow increase in load at that point leads to a decrease in bus voltage, reaching the ability of the power system to withstand load increase. If the critical point is exceeded, voltage instability will occur.

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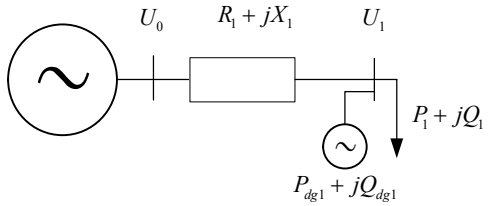


Figure 1. Single point distribution system with one DREs integration.

From Figure 1, when the integrated renewable energy is large, the power flows from Bus 1 to Bus 0, then the overvoltage problem occurs at Bus1. The analytical equation of the static voltage stability limit power can be expressed by [7]:

$$P_{dg1\max} = \frac{U_0^2 \cos\varphi}{2Z_1(1-\cos(\theta-\varphi))} + P_1 \tag{1}$$

where $P_{dg1\max}$ denotes the transmission power limit with static voltage stability as the constraint, φ denotes the load power factor angle, and θ is the impedance angle of the distribution line.

However, usually, the integrated renewable energy is less than $P_{dg1\max}$ owing to the reasonable operating range of voltage (usually $|\Delta V| \le 10\%$).

Assuming the DREs work at the maximum power point tracking point, which means the output reactive power is 0, the relationship between U_1 and P_{dg1} can be expressed by:

$$\begin{cases} U_1 = \sqrt{1+\left[P_1'^2+Q_1'^2\right]\left(R_1^2+X_1^2\right)-2\left[P_1'R_1+Q_1'X_1\right]} \\ P_1' = P_1 - P_{dg1} + ((P_1-P_{dg1})^2+Q_1^2)R_1 \\ Q_1' = Q_1 + ((P_1-P_{dg1})^2+Q_1^2)X_1 \end{cases} \tag{2}$$

From (2), the voltage of Bus 1 U_1 increases along with the increase of P_{dg1}. Therefore, if the operation range of bus voltage is δ, the maximum integration ratio of DREs in Figure 1 can be directly obtained by $\Delta U_1 = U_1 - 1 \le \delta$.

B. Overvoltage problem of the distribution system with two DREs

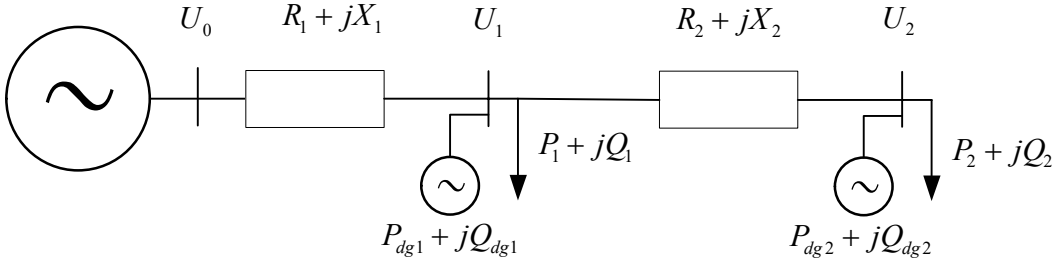


Figure 2. Single point distribution system with two DREs integration.

As shown in Figure 2, the relationship between the bus voltages (U_1 and U_2) and the integrated DREs (P_{dg1} and P_{dg2}) can be expressed by:

$$\begin{cases} U_1 = \sqrt{\left[1-\left(a^2+b^2\right)\left(R_1^2+X_1^2\right)+aR_1+bX_1\right]^2+\left[aX_1-bR_1\right]^2} \\ a = \left[\left(P_2-P_{dg2}\right)^2+Q_2^2\right]R_2+P_1+P_2-S_{DERs} \\ b = \left[\left(P_2-P_{dg2}\right)^2+Q_2^2\right]X_2+Q_1+Q_2 \end{cases} \tag{3}$$

where S_{DERs} is the total DREs amount. In (3), $S_{DERs} = P_{dg1} + P_{dg2}$.

The optimal allocation of the integrated DREs (P_{dg1} and P_{dg2}) is to achieve the minimum overvoltage of the bus voltages (U_1 and U_2). From (3), we can get the conclusion that if $P_2 = P_{dg2}$, then the parameters (a and b) achieve the minimum value. If $X_1 = R_1$, then U_1 achieves the minimum value. In the distribution power system, the line resistance and reactance are relatively close, so the optimal allocation of DREs in Figure 2 is $P_2 = P_{dg2}$.

C. Overvoltage problem of the distribution system with N DREs

As shown in Figure 3, the distribution system containing N nodes with a radial structure can be divided into two parts: one is the last power load node, and the remaining part can be seen as an equivalent source part. Therefore, the theory of the distribution system with two DREs integration is applicable to that with N DREs.

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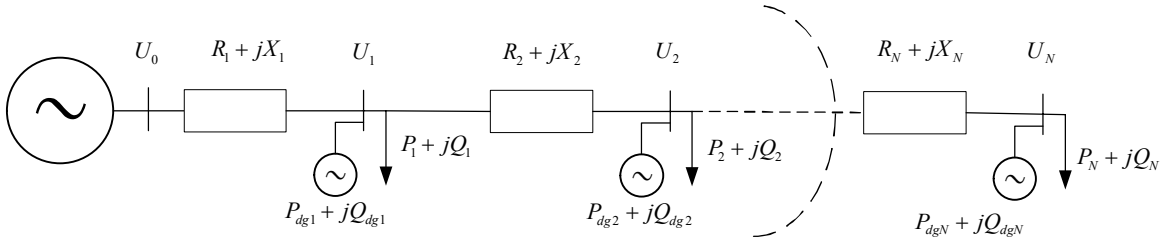


Figure 3. Single point distribution system with N DREs integration.

In sum, the optimal allocation method of DREs considering static voltage stability as the constraint can be expressed by:

$$\begin{cases} P_{dgN} = P_N \\ P_{dgN-1} = P_{N-1} \\ \vdots \\ P_{dgk} = S_{DREs} - \sum_{j=0}^{k-1} P_{N-k} \quad k \geq 1 \end{cases} \tag{4}$$

where S_{DREs} is the total amount of the DREs.

If k=1, the node closest to the source node (the substation) takes the remaining DREs allocation, and the static voltage stability of the distribution system with a high proportion of DREs is optimal.

III. The maximum integration ratio of DREs in the distribution power system

As shown in Figure 4, through the above theoretical analysis, the maximum integration ratio of DREs in a distribution system with static voltage stability as a constraint can be obtained by the following steps:

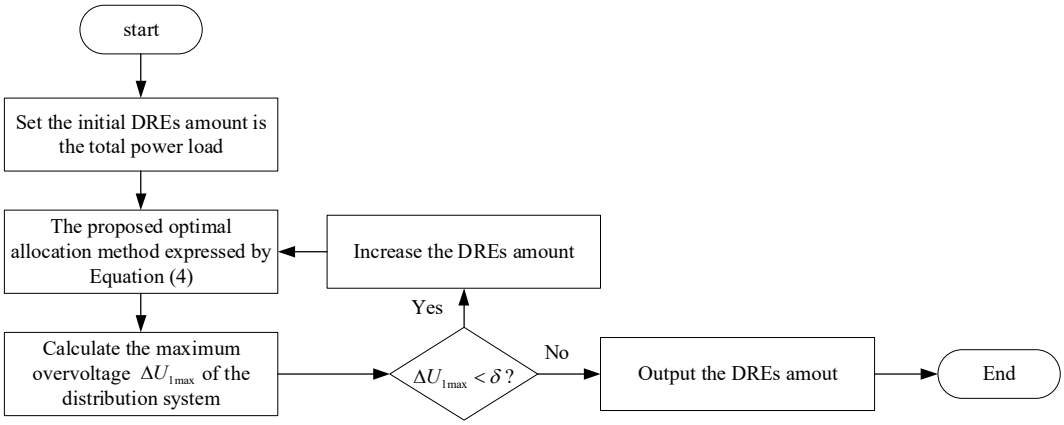


Figure 4. The maximum integration ratio of DREs in the distribution power system.

- Step 1: The maximum integration ratio of DREs in the distribution system is always subjected to the overvoltage problem, which means the DREs amount is larger than the total power load. So in the first step, the initial DREs amount S_{DREs} is set to equal to the total power load.
- Step 2: The DREs amount S_{DREs} is allocated according to the proposed optimal allocation method by (4).
- Step 3: The maximum overvoltage of the distribution power system is calculated. According to (4), the voltage of the node closest to the source node (the substation) is the highest voltage, so this step just calculates the voltage of Bus 1 $U_{1\max}$.
- Step 4: The maximum overvoltage $\Delta U_{1\max} = U_{1\max} - 1$ is compared with the operation range δ. If $\Delta U_{1\max} < \delta$. so, it is necessary to go to Step 5. If $\Delta U_{1\max} = \delta$, then it is recommended to go to step 6.
- Step 5: It is suggested to increase the DREs amount S_{DREs}, and go to Step 2.
- Step 6: $\Delta U_{1\max} = \delta$ means the maximum integration ratio of DREs in the distribution system with the static voltage stability as a constraint is obtained, then the DREs amount S_{DREs} is output.

IV. Case study

This section takes a distribution power system consisting of five DREs as an example to validate the effectiveness of the

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proposed method (N=5 in Figure 3). The parameters are listed in Table 1.

Table 1. The Main Parameters.

Parameters	Value	Parameters	Value
Resistance of Line	0.21 Ω/km	The Power Load at Bus 2	2 MW
Reactor of Line	0.398 Ω/km	The Power Load at Bus 3	3 MW
Line Length	1 km	The Power Load at Bus 4	6 MW
The Transformer Capacity	50 MVA	The Power Load at Bus 5	4 MW
Voltage of the Substation	10 kV	The Operation Range of Voltage	7%
The Power Load at Bus 1	5 MW	Power Factor of Power Load	1

It is assumed that the total amount of DREs is 60 MW, and four cases are studied in this section.

- Case 1: All the DREs are allocated at Bus 2, which means $P_{dg2} = 60$ MW.
- Case 2: The DREs are allocated at the five buses equally, which means $P_{dgi} = 12$ MW, $i = 1 \sim 5$.
- Case 3: The DREs are allocated proportional to their respective loads, which means $P_{dgi} / P_i = \frac{S_{DREs}}{\sum_{i=1}^{5} P_i} = \frac{60}{40} = 1.5$, $i = 1 \sim 5$.
- Case 4: The DREs are allocated by the proposed method, which means $P_{dg1} = 45$ MW, $P_{dg2} = 2$ MW, $P_{dg3} = 3$ MW, $P_{dg4} = 6$ MW, and $P_{dg5} = 4$ MW.

The voltage distribution among the five nodes is shown in Figure 5.

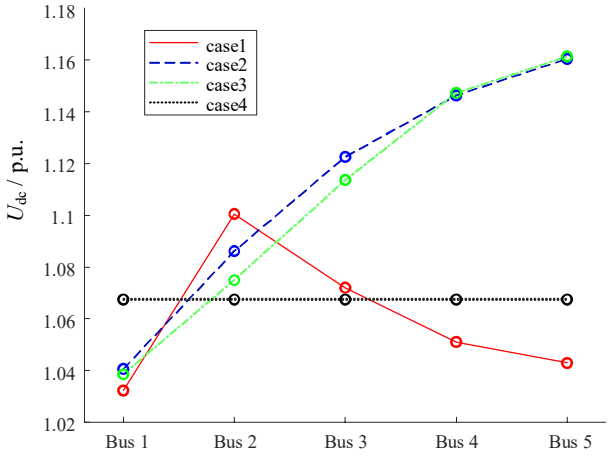


Figure 5. The comparison of voltage distribution with different allocation methods.

From Figure 5, the proposed allocation method of DREs can achieve a safe operation voltage, and the voltages of the five nodes are almost equal, which can validate the maximum amount of DREs that can be achieved by the proposed optimal allocation method. In this case, the maximum integration amount of DREs with static voltage stability as a constraint is 60 MW. Moreover, the reverse powers with different cases are shown in Figure 6, which shows that the proposed allocation method of DREs can achieve high efficiency and low power loss.

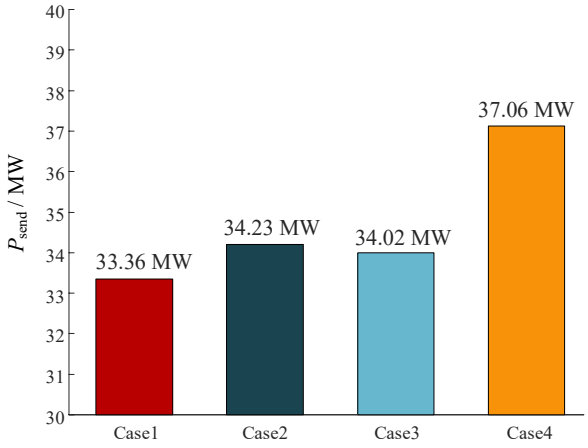


Figure 6. The comparison of reverse power with different allocation methods.

V. Conclusion and future work

The allocation value of DREs in distribution with a radial network has a great influence on the static voltage of the distribution nodes. An unreasonable allocation value of DREs in distribution will cause some nodes to exceed the static voltage limits and result in a large power transmission loss. Hence, this paper proposes a novel optimal allocation method for DREs in a distribution power system with static voltage stability as the constraint. Based on the optimal allocation method, one maximum integration ratio of DREs in the distribution power system is studied. Through the theoretical analysis and simulation verification, it can be concluded that the optimal allocation value of DREs should be equal to the power load except for the node closest to the source node (the substation). Compared to other allocation methods of DREs in distribution, this allocation method can avoid power backflow except for the node closest to the source node, thereby preventing some remote nodes from exceeding the voltage limit and reducing the power losses caused by line impedance during the transmission.

The impact of the reactive power on the allocation method suitable for the distribution with a ring network will be studied in further work.

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References

[1] Tanmay Das, Ranjit Roy, and Kamal Krishna Mandal. (2020) Impact of the penetration of distributed generation on optimal reactive power dispatch. Protection and Control of Modern Power Systems, 5(4): 332-357.

[2] H Bevrani, A Ghosh, and G Ledwich. (2010) Renewable energy sources and frequency regulation: survey and new perspectives. IET Renewable Power Generation, vol. 4, pp. 438-457.

[3] J Seuss, M J Reno, R J Broderick, and S Grijalva. (2015) Improving distribution network PV hosting capacity via smart inverter reactive power support. In Proc. 2015 IEEE PES General Meeting, Denver, Colorado, USA, pp. 1-5.

[4] Li Y, Liu H, and Fan Y, et al. (2020) Research on the Technology Requirement of Distributed Renewable Energy AC/DC Output System and Network Voltage Coordinated Control. In Proc. the 4th IEEE Conference on Energy Internet and Energy System Integration, Wuhan, China, pp. 1-4.

[5] S Ranjan and A R Abhyankar. (2019) Network Constraint Compliant Choice of Renewable Energy Source Sitting. In Proc. 2019 8th International Conference on Power Systems (ICPS), Wuhan, China, pp. 1-4.

[6] A F Nasef, H A-H Khattab, and D S M Osheba. (2019) Optimal Allocation of Renewable Energy Sources in Electrical Power System Using Oscillatory Particle Swarm Optimization. In proc. 2019 IEEE Conference on Power Electronics and Renewable Energy (CPERE), Aswan, Egypt, pp. 1-6.

[7] Xu S, Zhou, B, and Peng L. (2023) Study on margin ratio index of static voltage stability of multiple renewable energy stations delivery system. Power System Technology, https://kns.cnki.net/kcms/detail/11.2410.TM.20230506.1150.009.html.

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